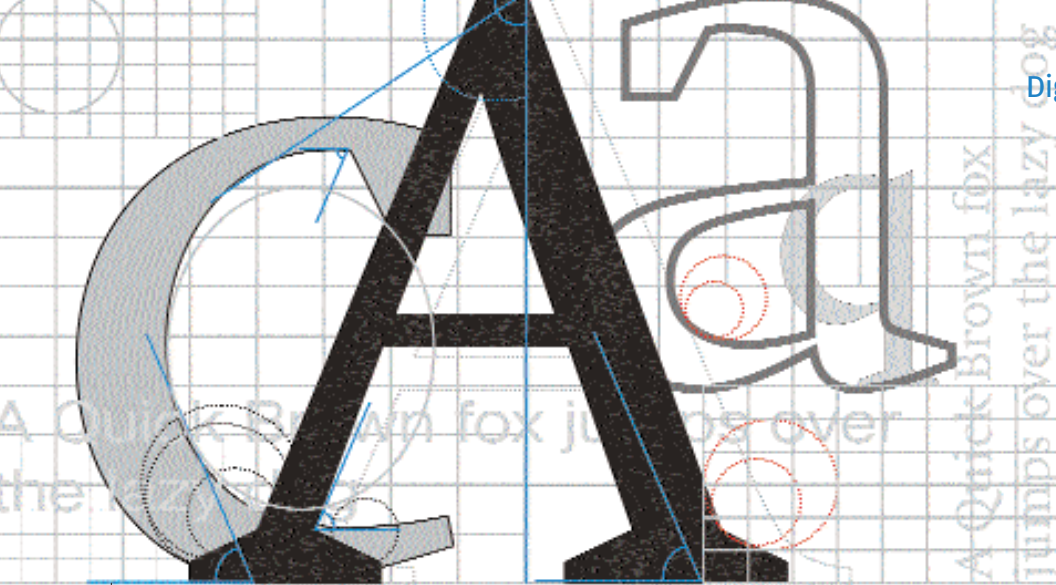




Imaging Rohit A Chandwaskar

Fonts

How does text emerge in pretty

Samir Makwana

Fonts make possible the text of the magazine you're reading, and the text that appears on your computer screen, whatever the application and OS. Without them, we wouldn't have an issue to print every month.

Fonts are sets of *glyphs*—a glyph is a design or graphical shape given to characters—letters, numbers, symbols, ideograms, and punctuation marks. Seen another way, a font is a collection of typefaces along with characteristics like size, pitch, and spacing, along with the “rasters” for rendering the text. (A geometric data set with values assigned to a collection of pixels or dots is a raster.)

A typeface is the primary aspect of a font: it is a set of characters with a particular style. The basic difference between a typeface and a font is that a font is a character set with one typeface, design, and size, while a typeface is just a set of characters with the same style.

A font family is a group of fonts that exhibit similarities in design and that are to be used together. Different styles—as in size, weight, italics, bold, etc. lend variation to a font family. To take an everyday example, Arial is the font, whereas Arial Italic and Arial Bold are fonts in the Arial family.

Font Formats

Font files can be classified by formats as in the following.

Bitmap Fonts: These consist of a matrix of pixels representing glyphs. They are used for printing or screen display. Bitmap screen fonts are used in Linux consoles, Windows recovery consoles, and text editors, where there is lack of scalability. The font encoding for bitmap fonts is inferior to that of other fonts, so they lack in visual quality and flexibility. Being device-dependent, they can only be used at a particular resolution; a 100 dpi bitmap font will be displayed as 100 dpi even on a 1200 dpi monitor.

Vector Graphics Format: Vector Graphics define geometrically the shape or outline of a font. A font in this format contains a collection of vector images of glyphs. In vector images, geometrical formulas are used to represent images in computer graphics, so they can be stretched and resized. These fonts can be scaled to any size, which requires numerical processing—each font description represents innumerable characters. For example, Verdana 10-point is one font, and Verdana 14-point is another. PostScript and TrueType are popular “languages” for defining such “scalable” fonts.

PostScript Fonts: For electronic and desktop publishing, PostScript, a page description and programming language, is used.

When the need for a common standard to integrate the GUI between the first Macintosh and laser printers arose, Adobe Systems came out with scalable Type 1 fonts in PostScript, which would render fonts (generate a display/image using computer programs) for both printer and screen.

TrueType Fonts: These were designed and developed by Apple, and are now used by both Apple's Mac OS X and Microsoft's Windows. The TrueType fonts that appear on the screen or on paper contain their own algorithms for conversion of text into pixels. For other types of fonts, a rasteriser is required: this is software that converts text into pixels, and comes with operating systems.

Rendering With Applications

Every font file has information in the form of mathematical data (basically, codes) assigned to each character. Since variations can be applied to fonts, it would become difficult to assign numbers to each character. So, standards such as Unicode, for example, where each character has a unique number on any platform, come into the picture.

For a particular font, a table is computed in which information is mapped about what glyph to use for a particular character code. (These tables vary according to computing platform.) Mapping from Unicode (for example) to glyphs, an index of glyphs is created. The application will render the respective glyph from the index according to the text, and the corresponding character is displayed on the screen.

The differences between vector and TrueType fonts lie in the way the glyph for each character is stored in the font file. In vector fonts, a glyph is a collection of line end-points that define the line segments Windows uses to draw a character in the font. In TrueType fonts, a glyph is a collection of line and curve commands as well as a collection of hints. “Hints” refers to information included in a font file to aid the rasteriser in scaling character outlines to various sizes.

As you can see, there's a lot more to be learnt about fonts—a good place to start is www.microsoft.com/typography/default.mspx■

samir_makwana@thinkdigit.com